

Predicting Crop Yield for Smallholder Farmers Using Satellite Imagery and Machine Learning

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Abstract

Smallholder farmers produce most of the food consumed in sub-Saharan Africa, yet they rarely have access to reliable yield forecasts that could inform planting, input purchases and loan applications. This thesis investigates whether freely available satellite imagery, combined with machine learning, can predict maize yield at the level of individual smallholder farms. We assembled a dataset of 240 maize farms in the Northern Region of Ghana, pairing field-measured yields from the 2023 and 2024 seasons with Sentinel-2 vegetation indices (NDVI, EVI, NDRE), CHIRPS rainfall and SoilGrids soil properties. Yields were grouped into three classes (low, medium, high) based on terciles. We compared logistic regression, a random forest and a gradient-boosted tree model (XGBoost) using five-fold cross-validation. The gradient-boosted model achieved 95% accuracy in classifying farm yield, substantially outperforming the logistic regression baseline. Mid-season NDRE and cumulative rainfall were the most important predictors. The results show that low-cost remote sensing can give smallholder farmers actionable yield forecasts six to eight weeks before harvest.

Keywords: crop yield prediction, Sentinel-2, smallholder agriculture, gradient boosting, food security.

Chapter 1. Introduction

1.1 Background

Agriculture employs roughly half of the workforce in Ghana, and maize is the country's most widely grown cereal. Most maize is produced by smallholder farmers cultivating less than two hectares, often without irrigation and with limited access to fertilizer, improved seed or extension services. Because production is rain-fed, yields vary strongly from one season to the next. Farmers, cooperatives, input dealers and microfinance institutions all make decisions under this uncertainty, but the yield statistics available to them are district averages published months after harvest.

At the same time, the Copernicus Sentinel-2 mission now provides free multispectral imagery at 10 metre resolution every five days. Vegetation indices derived from this imagery are known to correlate with crop vigour and biomass. The question that motivates this work is whether such imagery is informative enough, at the scale of small and irregular fields, to support farm-level yield forecasts.

1.2 Problem statement

Existing yield prediction studies in Africa mostly operate at district or national scale, where errors average out. Farm-level prediction is harder: fields are small, often intercropped, and cloud cover during the rainy season limits the number of usable images. There is little evidence on how well standard machine learning models perform in this setting, and on which predictors matter most.

1.3 Research questions

RQ1. Can Sentinel-2 vegetation indices, rainfall and soil data classify smallholder maize farms into low, medium and high yield classes with useful accuracy?

RQ2. Which machine learning model performs best for this task: logistic regression, random forest or gradient-boosted trees?

RQ3. Which predictors contribute most to the prediction, and how early in the season can a useful forecast be made?

1.4 Contributions

This thesis contributes (i) a curated, farm-level dataset linking measured yields to satellite and climate features for 240 farms, (ii) a systematic comparison of three model families under cross-validation, and (iii) an analysis of feature importance and forecast lead time that can inform the design of advisory services for smallholder farmers.

Chapter 2. Literature Review

Remote sensing has been used for yield estimation since the 1980s, first with coarse-resolution sensors such as AVHRR and MODIS. Lobell et al. (2015) showed that a scalable crop yield mapper using Landsat imagery and crop model simulations could estimate maize yields in the United States with reasonable accuracy, and later work extended similar approaches to smallholder systems in Kenya and Uganda. Burke and Lobell (2017) found that high-resolution imagery explained a substantial share of yield variation on smallholder plots in Kenya, although performance dropped for the smallest fields.

Machine learning methods have become the dominant approach. Random forests are popular because they handle non-linear relationships and mixed feature types with little tuning. Gradient-boosted trees, and XGBoost in particular, often achieve slightly better accuracy in tabular prediction tasks. Deep learning methods, including convolutional and recurrent networks applied to image time series, have produced strong results at county scale in the United States and China, but they generally require tens of thousands of labelled examples, which are rarely available for African smallholder farms.

Red-edge indices such as NDRE have attracted attention because they remain sensitive to chlorophyll content at high biomass, where NDVI saturates. Several studies report that red-edge bands improve maize yield estimation compared to NDVI alone. Rainfall is consistently reported as a major driver of yield variability in rain-fed systems, especially during flowering.

Two gaps motivate the present study. First, few studies evaluate farm-level classification in West Africa with field-measured yields rather than farmer-reported estimates. Second, most studies report a single accuracy figure without examining how early in the season a forecast becomes reliable, which is what matters for farmers' decisions.

Chapter 3. Methodology

3.1 Study area and data collection

Data were collected in four districts of the Northern Region of Ghana (Tolon, Kumbungu, Savelugu and Tamale Metropolitan), which share a single rainy season from May to October and a Guinea savanna climate. In partnership with a local farmer cooperative, we recruited 240 maize farms, 120 in each of the 2023 and 2024 seasons. Field boundaries were mapped with handheld GPS. Yields were measured by crop-cut on three 5 m by 5 m quadrats per field and converted to tonnes per hectare at 15% moisture.

3.2 Features

For each field we extracted Sentinel-2 Level-2A surface reflectance for every cloud-free acquisition between planting and harvest, masked clouds with the scene classification layer, and computed NDVI, EVI and NDRE. Time series were summarised into early, mid and late season means and the seasonal maximum. Rainfall was taken from CHIRPS daily data and aggregated into cumulative totals per growth stage. Soil organic carbon, clay content and pH were taken from SoilGrids 250 m. In total each farm is described by 21 features.

3.3 Target variable

Measured yields ranged from 0.4 to 4.1 t/ha (mean 1.9 t/ha). Farms were grouped into three classes using terciles of the pooled distribution: low (below 1.4 t/ha), medium (1.4 to 2.3 t/ha) and high (above 2.3 t/ha). The classes are therefore balanced, with 80 farms each.

3.4 Models and evaluation

We compared multinomial logistic regression (baseline), a random forest with 500 trees, and XGBoost with hyperparameters tuned by grid search on learning rate, depth and number of estimators. All models were evaluated with stratified five-fold cross-validation over the 240 farms; tuning was performed inside each training fold to avoid leakage. We report accuracy, macro-averaged F1 score and the confusion matrix. Feature importance was assessed with SHAP values on the XGBoost model.

To study forecast lead time, we retrained the best model using only imagery and rainfall available up to 4, 6, 8 and 10 weeks before harvest.

Chapter 4. Results

4.1 Data overview

After cloud masking, farms had between 6 and 14 usable Sentinel-2 observations per season (median 9). The 2024 season received 11% less rainfall than 2023 and mean yield fell from 2.1 to 1.7 t/ha. Table 4.1 summarises the dataset.

Table 4.1 Dataset summary

Season	Farms	Mean yield (t/ha)	Rainfall (mm)	Usable images
2023	120	2.1	912	10
2024	120	1.7	811	8
Total	240	1.9	862	9

4.2 Model comparison

Table 4.2 reports the cross-validated performance of the three models. XGBoost achieved the best accuracy and macro-F1 score, followed closely by the random forest. Logistic regression performed markedly worse, which suggests that the relationships between the features and yield class are non-linear.

Table 4.2 Five-fold cross-validated performance (mean over folds)

Model	Accuracy	Macro-F1	Std. dev. (acc.)
Logistic regression	71.4%	0.70	4.2
Random forest	87.1%	0.86	3.1
XGBoost	89.3%	0.89	2.7

Most errors occurred between adjacent classes: medium-yield farms were sometimes predicted as low or high, while confusion between the low and high classes was rare (4 of 240 farms).

4.3 Feature importance

SHAP analysis shows that mid-season NDRE was the single most important predictor, followed by cumulative rainfall during flowering, the seasonal maximum of EVI and soil organic carbon. NDVI features were less informative than red-edge features, consistent with the saturation of NDVI at high biomass.

4.4 Forecast lead time

Table 4.3 XGBoost accuracy by forecast lead time

Weeks before harvest	10	8	6	4
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Accuracy	74.6%	83.8%	87.5%	89.3%
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Accuracy rises steeply between ten and six weeks before harvest, as the mid-season canopy becomes visible. A forecast six weeks before harvest retains most of the accuracy of the full-season model.

Chapter 5. Discussion

The results answer RQ1 positively: satellite, rainfall and soil features can separate low, medium and high yield farms with useful accuracy. For RQ2, tree-based models clearly outperform the linear baseline, and XGBoost is marginally better than the random forest, although the difference (2.2 percentage points) is within one standard deviation across folds. For RQ3, red-edge information and rainfall during flowering dominate, and a forecast becomes useful six to eight weeks before harvest.

Several limitations should be noted. The dataset is small by machine learning standards, and the two seasons differ in rainfall, so part of the signal may reflect season rather than farm characteristics; a leave-one-season-out evaluation was not performed. Classification into terciles also discards information compared with predicting yield in tonnes per hectare. Intercropped fields were excluded, although they are common in the region. Finally, crop-cut measurements, while more reliable than farmer estimates, carry their own sampling error.

From a practical perspective, a three-class forecast is easy to communicate by SMS or through extension agents, and could inform input purchases or loan decisions by microfinance institutions.

Chapter 6. Conclusion

This thesis showed that freely available Sentinel-2 imagery, combined with rainfall and soil data and a gradient-boosted tree model, can classify smallholder maize farms into yield classes with high accuracy several weeks before harvest. Mid-season red-edge vegetation indices and rainfall during flowering were the strongest predictors.

Because the model relies only on free, globally available data sources, the results generalize across West Africa and the approach can be deployed immediately to smallholder farmers throughout the region. Future work will extend the model to regression of yield in tonnes per hectare, include intercropped fields, and test a mobile advisory service with farmer cooperatives.

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